

# General Requirements for the Group Project

## I.1. Requirements for Project Implementation and Reporting

- The report must be prepared using a word processor, with font size 12 and line spacing of 1.4. Page margins shall be set as follows: top margin 2.5 cm, bottom margin 2.5 cm, left margin 3.5 cm, and right margin 2.0 cm. The length of the report must be between 15 and 25 pages, excluding the references section.
- All figures and illustrations used in the theoretical sections must be independently recreated by the students (redrawn or newly generated). Direct reproduction of figures from external sources is not permitted.
- All mathematical expressions must be typeset using appropriate equation editors (such as LaTeX, Equation Editor, etc.). The use of screenshots or image-based representations of formulas is strictly prohibited.
- Any form of plagiarism, including copying content from documents, online sources, or reports from other groups without proper and valid citation, is strictly forbidden. Any violation will result in a score of zero (0) for all members of the group.
- The content presented in the report must ensure that all group members fully understand the material and are able to explain it competently when requested.
- All group members are required to be present at the presentation session.

## I.2. Assessment Criteria

The major assignment will be evaluated based on the following components:

- **Report format, submission deadline, and preparation for the presentation:** 30%  
This component includes timely submission, punctual attendance, and adequate preparation of presentation materials, programs, datasets, and all necessary equipment required for the presentation.
- **Oral presentation (approximately 10 minutes):** 30%  
Evaluation is based on clarity of presentation, logical organization, effective communication skills, and appropriate emphasis on the key aspects of the project.
- **Level of understanding of group members and overall project quality:** 40%  
Assessment is based on mastery of the subject matter, the ability to respond accurately to questions, and the depth of understanding demonstrated by the group members regarding the implemented topic.

## I.3. Structure of a Major Assignment Project

### 1. Cover Page

- Name of the university and faculty/department
- Course title
- Project title

- Instructor
- Student group information (full names, student IDs, class)
- Description of the contribution of each group member

## 2. Table of Contents

## 3. Introduction

This section provides the background and overall orientation of the project, enabling readers to understand its context and motivation.

### Main content includes:

- **Motivation**  
Present the scientific significance and/or practical relevance of the topic. Students should clearly address:
  - The importance of the problem from a mathematical or applied perspective;
  - Relevant application domains (e.g., engineering, economics, machine learning, image processing, etc.);
  - Real-world problems that the project aims to address.
- **Objectives of the project**  
Clearly state the specific outcomes to be achieved, such as:
  - Presenting the theoretical foundations of the method;
  - Developing an algorithm;
  - Implementing a computational program;
  - Applying the method to an illustrative example.
- **Scope of the study**  
Clearly define the boundaries of the project, including:
  - Research objects (e.g., types of matrices, datasets, models);
  - The balance between theoretical analysis and practical application;
  - Topics and aspects not covered in the study.

## 4. Theoretical Background

This section presents the mathematical foundations underlying the method.

- Required definitions (with explanations of their meanings);
- Relevant theorems (including their roles and significance within the project).

Mathematical formulas should not merely be listed; their mathematical interpretation and relevance must be clearly explained.

## 5. Algorithm or Mathematical Model Employed

This section serves as the link between theory and implementation.

### Main content includes:

- Description of the algorithmic steps in mathematical form, derived from the theoretical background;
- Explanation of the theoretical basis for each step;
- If an applied model is used, presentation of the model in the form of mathematical equations;
- Complexity analysis, where appropriate.

## 6. Results, Deliverables, and Experimental Evaluation

This section presents all outputs of the project along with validation and evaluation results.

- **Experimental results**
  - Illustrative examples;
  - Program execution results;
  - Input–output comparisons;
  - Analysis and discussion of the results.
- **Project deliverables**
  - Source code;
  - Executable files (if applicable);
  - Output data in various formats, including:
    - Images (charts, graphs, visualizations);
    - Audio files (for signal processing topics);
    - Data files (.csv, .txt, .mat, .json, etc.);
    - Simulation or demonstration videos (if available).
- **Technical description**
  - Programming languages used;
  - Main libraries and tools;
  - Description of key functions or modules;
  - Explanation of critical code segments.

## 7. Conclusion and Future Work

Summarize the results achieved.

- Strengths and limitations of the proposed method;
- Comparison with alternative methods, where applicable;
- Suggested directions for future research or development.

## 8. References

List all references in accordance with standard academic citation formats, including textbooks, journal articles, and reputable scholarly websites.